chain from a free 3′OH end, but none can initiate synthesis.

- Synthesis of DNA requires nucleoside triphosphates or nucleotides - deoxyadenosinetriphosphate (dATP), deoxythymidinetriphosphate (dTTP), deoxycytidinetriphosphate (dCTP), deoxyguanosinetriphosphate (dGTP). When a nucleoside triphosphate bonds to sugar in a growing DNA strand, it loses two phosphates.
- RNA primer synthesised during initiation is removed and replaced with DNA by DNA polymerase I
- Sealing of the nick by DNA ligase which catalyses the formation of phosphodiester bond between a 3′ hydroxyl end of one DNA fragment and 5′ phosphate at the end of another strand.

The elongation phase of replication includes two distinct but related operations

- Leading Strand Synthesis
- Lagging Strand Synthesis

Leading strand synthesis begins with the synthesis of short (10 to 60 nucleotide long) RNA primer at the replication origin. Deoxyribonucleotides are added to this primer by DNA polymerase III. Leading strand synthesis then proceeds continuosly, keeping pace with the unwinding of DNA at the replication fork. The continuous strand or leading strand is one in which 5′→3′ synthesis proceeds in the same direction as replication fork movement.

Lagging strand or discontinuous strand is one in which 5′→3′ DNA synthesis proceeds in the direction opposite to the direction of fork movement. This strand is synthesised as short fragments, known as Okazaki fragments named after R. Okazaki. Okazaki fragments range in length from a few hundred to a few thousand nucleotides depending on the cell types. Each okazaki fragment must be initiated by the action of primase. Once an Okazaki fragment has been completed its RNA primer is removed and replaced with DNA by DNA polymerase I and the nick is sealed by DNA ligase.

The fidelity of DNA replication is maintained by (1) base selection by the polymerase, (2) a 3′→5′ proofreading exonuclease activity that is part of most DNA polymerases, and (3) specific repair systems for mismatches left behind after replication.

Infobits

RNA dependent DNA polymerases, also called reverse transcriptases, were first discovered in retroviruses, which convert their RNA genomes into double-stranded DNA as part of their life cycle. These enzymes transcribe the viral RNA into DNA, a process that can be used experimentally to form complementary DNA.

Termination

Eventually, the two replication forks of the circular *E.coli* chromosome meet at a terminus region called Ter (for terminus). The Ter sequence function as binding sites for protein Tus (Termination utilization substance). The Tus-Ter complex can arrest a replication fork from only one direction. When either replication fork encounters a functional Tus-Ter complex, it halts. The other fork halts when it meets the first (arrested) fork (Figure 14.15).

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